

PREPRINT

Addressing Geographic Access Disparities Under Time-Sensitive Logistics in Kidney Transplantation

Elijah Pivo

Institute for Data, Systems, and Society, Massachusetts Institute of Technology, Cambridge, MA 02139, epivo@mit.edu

Thomas Athey

Computer Science and Artificial Intelligence Laboratory, Massachusetts Institute of Technology, Cambridge, MA 02139, tathey_1@mit.edu

Nikolaos Trichakis, Dimitris Bertsimas

Sloan School of Management, Massachusetts Institute of Technology, Cambridge, MA 02139, ntrichakis@mit.edu, dbertsim@mit.edu

Recent reforms to U.S. kidney allocation policy have sought to reduce longstanding geographic disparities in access to transplantation by diminishing the importance of geographic proximity between kidney donors and recipients. As distances between donors and recipients have grown and logistical complexity has increased, rates of kidney nonuse, delayed graft function, and out-of-sequence allocation have risen as well, while geographic disparities persist. This has raised the question of whether geographic disparities can be eliminated without imposing impractical logistical requirements. In this work, we develop and validate a national-scale model of deceased-donor kidney transportation and use it to evaluate whether optimizing, rather than diminishing, the role of geographic proximity in kidney allocation policy can reduce geographic disparities while maintaining logistical feasibility. We retrospectively reallocate the kidneys transplanted in 2024 among that year’s transplant recipients, preserving the historical geographic distribution of transplants, and estimate that total transportation time could be reduced by 31.7%, air freight usage by 38.9%, and shipments longer than 10 hours by 66.7%, without substantially increasing waiting time or worsening kidney quality for any recipient. We then reallocate these kidneys among all compatible candidates to consider alternative geographic distributions of transplants. We find that regional transplant counts could have been brought within 3.5% of expected levels while still reducing transportation relative to the historical allocation. Our findings suggest that longstanding geographic inequities can be substantially reduced without imposing impractical logistical requirements.

Key words: kidney transplantation, organ allocation, geographic equity, healthcare logistics, transportation modeling, resource allocation

1. Introduction

In 2025, over 50,000 patients registered for a deceased-donor kidney transplant in the United States (OPTN 2026a). Transplantation can free patients from the burdens of chronic dialysis, restore quality of life, and dramatically extend life expectancy (Wolfe et al. 1999), but transplantation is also logistically challenging. While outside the body, a kidney accrues ischemic injury, which steadily increases the risk of complications (Siedlecki et al. 2011). Within only 24 hours of organ recovery, kidney function must be assessed, a recipient must be identified, and the kidney must be transported from donor to recipient. Beyond this time window, kidneys become increasingly likely to go unused, even amid a severe shortage.

Central to managing these logistical constraints is kidney allocation policy. This policy is set by the Organ Procurement and Transplantation Network (OPTN) and determines the priority of compatible transplant candidates when a kidney is donated. To limit logistical complexity, allocation policy prioritizes candidates based in part on their geographic proximity to the donor (OPTN 2025). However, geographic prioritization may contribute to inequities in patient access to transplantation, which varies substantially by candidate location in the United States. For example, Zhou et al. (2018) reported that similar candidates had a median 1.81-fold difference in transplantation rates across different donation service areas.¹ In recent years, policymakers have sought to reduce these geographic disparities.

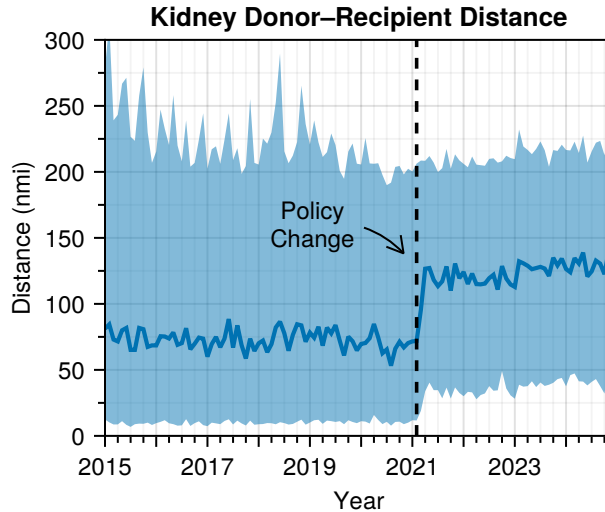


Figure 1 (Color online) Monthly Donor-Recipient Distance for Kidney Transplants, 2015–2024 (Median and Interquartile Range)

¹ Donation service areas are geographic regions in which a particular organ procurement organization coordinates organ donation. Historically, candidates within a donor’s donation service area were prioritized above those outside the donation service area.

A natural approach is to reduce the role of geographic proximity in candidate priority. This has been the direction pursued by the OPTN, which on March 15, 2021, began to use larger donor-centered circular zones to prioritize candidates rather than donation service areas (OPTN 2019, 2026b). This shift toward more location-agnostic allocation has come with substantial consequences. The distance between transplant recipients and their donors has grown sharply (Figure 1). Median donor–recipient distance rose from 73 to 124 nmi (+69.9%), while the 25th percentile increased from 10 to 37 nmi (+270.0%). Longer transportation distances, together with longer organ placement times,² increase the ischemic time experienced by donated kidneys. This may have contributed to several concerning trends in the kidney transplant system, including rising rates of kidney nonuse, delayed graft function, and out-of-sequence allocation (Lentine et al. 2025).³

Yet even as kidneys have been distributed more broadly, geographic disparities have persisted. For example, in 2024, several years after the policy change, the average waiting time for a kidney transplant was still nearly 6 years in San Francisco and 2 years in Cincinnati. This raises the question of whether it is possible to eliminate geographic disparities without imposing impractical logistical requirements.

Geographic disparities in kidney allocation may be addressed more efficiently in two respects. First, location-agnostic policy allows more distant matches regardless of whether they are necessary to address geographic disparities. In fact, under increasingly location-agnostic policy, donor–recipient distances increasingly resemble those between randomly selected individuals in the United States. By instead accounting for candidate location together with the geographic distributions of kidney supply and demand, more distant matches could be prioritized only where necessary to balance supply and demand. Consequently, the transportation required to achieve geographic equity could reflect the much smaller geographic mismatch between supply and demand, rather than by the geographic distribution of the population itself.

Second, current allocation policy measures proximity through geographic distance, which does not directly capture the primary logistical constraint: transportation time. Transportation time can vary widely for similar distances based on factors such as airport accessibility and flight availability. Accounting for transportation time directly can identify opportunities to more efficiently equalize access.

In this work, we investigate both approaches by optimizing donor–recipient assignments while explicitly modeling the transportation time associated with each potential match. To estimate

² Organ placement times are also impacted by the shift toward location-agnostic allocation, because offers for more distant kidneys are declined more frequently.

³ Kidney nonuse refers to the donation of a kidney that is ultimately not transplanted. Delayed graft function occurs when a transplanted kidney fails to sufficiently filter the recipient’s blood for a period after transplantation, requiring continued dialysis. Out-of-sequence allocation occurs when a kidney is offered to a lower-priority candidate before higher-priority candidates have been given the opportunity to accept it.

transportation times, we develop the first national-scale model of kidney transportation in the United States. The model incorporates road network geometry, historical flight schedules, and air cargo station operating hours. A specialized variant of the A* routing algorithm enables route planning in milliseconds, making it possible to incorporate transportation time directly into large-scale allocation analyses. We validate the model against historical logistics data from the OPTN.

Next, we use this model to assess whether the geographic distribution of kidneys following the 2021 policy change could have been achieved more efficiently. We reallocate the kidneys transplanted in 2024 among that year’s recipients to minimize transportation time while requiring that no recipient experiences a substantial increase in waiting time or decrease in kidney quality. We find that total transportation time could have been reduced by 31.7%, air freight usage by 38.9%, and the number of shipments longer than 10 hours by 66.7%.

However, reallocating kidneys among the same historical recipients necessarily preserves the geographic distribution of transplants observed in 2024, when regional transplant counts ranged from 37.4% below to 59.1% above waitlist-proportional levels (the number of transplants expected based on the candidates listed in each region). We therefore conduct a second analysis in which we relax the requirement that kidneys be allocated among historical recipients and instead allow allocation to any eligible candidate listed that year. We find that regional transplant counts could be brought within 3.5% of waitlist-proportional levels while preserving similar donor–recipient pairing characteristics at the national level and reducing total transportation time by 45.0%.

This work suggests a different perspective on the design of kidney allocation policy. Rather than asking how much the role of geographic proximity should be diminished, we ask whether it should instead be optimized. Our findings indicate that optimizing, rather than diminishing, the role of geographic proximity can substantially ease the longstanding tradeoff between geographic equity and logistical efficiency, opening an opportunity for future work on allocation policies that operationalize this approach. More generally, our findings illustrate a broader principle for the design of allocation systems: eliminating the effect of a characteristic on outcomes need not require eliminating that characteristic from the decision process. When ignoring that characteristic introduces secondary costs, explicitly accounting for it may allow the desired outcomes to be achieved more efficiently.

2. Literature Review

The complex challenges of organ transplantation have motivated a rich literature within management science and operations research. One particular area of focus has been techniques for evaluating and designing organ allocation policies. Simulation-based policy evaluation has played an important role in policy development in practice (Pritsker et al. 1995, 1996, Harper et al. 2000,

Thompson et al. 2004), while early analytical work employed stochastic sequential assignment problems and queuing-theoretic models to characterize waitlist dynamics and study allocation principles (David and Yechiali 1985, 1990, 1995, David 1995, Zenios 1999, Su and Zenios 2004, 2005). Subsequent work developed fluid approximations of queuing dynamics, often to allow for the use of optimal control techniques (Zenios et al. 2000, Akan et al. 2012, Ata et al. 2021). Optimization-based approaches for policy design (Bertsimas et al. 2013, Pivo et al. 2025) and structural models that account for strategic acceptance behavior have also been proposed (Agarwal et al. 2021). Our paper evaluates aspects of organ allocation policy, but by analyzing donor–recipient assignments directly avoids challenges in policy design that can obscure the potential for operational improvements.

Geographic disparities in patient access to transplantation and organ transportation have been of interest to researchers. A number of studies have optimized geographical regions used during organ allocation. Some have sought to balance transportation distances with the likelihood of successful organ allocation (Kong et al. 2010), while others have sought to reduce geographic disparity (Gentry et al. 2015, Kilambi and Mehrotra 2017, Akshat et al. 2023). Transportation has been examined as a mechanism for reducing geographic disparities, including proposals that rely on air transportation of transplant candidates (Ata et al. 2017), as well as empirical studies showing that improved airline connectivity facilitates organ sharing (Wang et al. 2022). Our work contributes to this literature by explicitly modeling the operational logistics of kidney transportation and optimizing transplant assignments at the patient level.

Other work has examined how information influences acceptance decisions, including the role of previous refusals (Zhang 2010) and whether publicly available waitlist information enables patients to make effective acceptance decisions (Sandıkçı et al. 2013). Dai et al. (2020) study how providing allocation priority to registered organ donors influences donor registration incentives, while Arıkan et al. (2018) considers effect of geographic allocation regions on organ recovery decisions. In addition, a substantial literature has studied living donor kidney exchange, where optimization methods have been used to coordinate exchanges among incompatible donor–recipient pairs (Roth et al. 2004, 2005, Ding et al. 2018, Agarwal et al. 2019).

Our paper is relevant to several streams of research beyond transplantation. First, it contributes to the growing literature on algorithmic fairness and decision-making in public systems showing that achieving fairness with respect to a characteristic may require explicitly incorporating that characteristic into decision-making (Dwork et al. 2012, Corbett-Davies et al. 2017, Kleinberg et al. 2018). Second, this paper contributes to the transportation and logistics literature on high-performance route planning in time-dependent transportation networks (Gendreau et al. 2015, Bast et al. 2016). Third, this paper contributes to the operations management literature on balancing efficiency and

equity in the allocation of scarce public resources, which has examined this tradeoff in contexts such as public facility location (Marsh and Schilling 1994), disaster relief and humanitarian logistics (Holguín-Veras et al. 2013), food bank distribution (Lien et al. 2014), and healthcare and vaccine distribution (Enayati and Özaltın 2020). Much of this literature treats equity and efficiency as fundamentally competing objectives; our findings suggest that in the kidney transplantation context, this tradeoff may be less severe than previously assumed.

3. Kidney Transportation Model

Section 3.1 describes standard practices for kidney transportation and the structure of our transportation model. Section 3.2 introduces the Lazy A* algorithm and compares its performance with standard route-planning algorithms. Finally, Section 3.3 validates the transportation model against historical data.

3.1. Transportation Practices and Model Structure

When a kidney is donated for transplantation, it must be transported from the donor hospital where the organ is recovered to the transplant center of the recipient. Most kidneys are transported by ground, with a specialized courier picking up the packaged kidney and driving it directly to the transplant center. For longer-distance shipments, kidneys may instead be transported as air freight in the cargo holds of commercial passenger flights. Other, less commonly used methods include accompanied air transport, in which a courier carries the packaged kidney aboard a commercial flight, and chartered air transport.

The transportation model estimates the route used to transport a kidney from a donor hospital to a transplant center at a particular departure time as a fastest-path through a directed graph $G = (V, E)$ with time-varying edge weights, where V is the set of vertices and E is the set of directed edges between the vertices. A diagram of the graph is shown in Figure 2. The vertices include a set of donor hospital vertices $V_{DH} \subset V$, a set of airport vertices $V_{AP} \subset V$, and a set of transplant center vertices $V_{TXC} \subset V$. Altogether, the vertices represent 3,133 donor hospitals, 130 transplant centers, and 373 airports. Locations of donor hospitals and transplant centers were obtained from OPTN data, while airport locations were obtained from the IP2Location IATA/ICAO airport database (IP2Location 2026), distributed under the CC BY-SA 4.0 license. Facility locations are mapped in Figure 3.

The weight of an edge $e = (v_1, v_2) \in E$ is denoted $w_e(t)$ and represents the travel time from vertex v_1 to v_2 when departing at time t . Ground transport is included in the graph through edges from each donor hospital vertex $v_{dh} \in V_{DH}$ to each transplant center vertex $v_{txc} \in V_{TXC}$ with edge weights $t_{drive}(v_{dh}, v_{txc})$. The dashed edge in Figure 2 is an example. Drive times were estimated with the Open Source Routing Machine (OSRM) software package (Luxen and Vetter 2011). OSRM

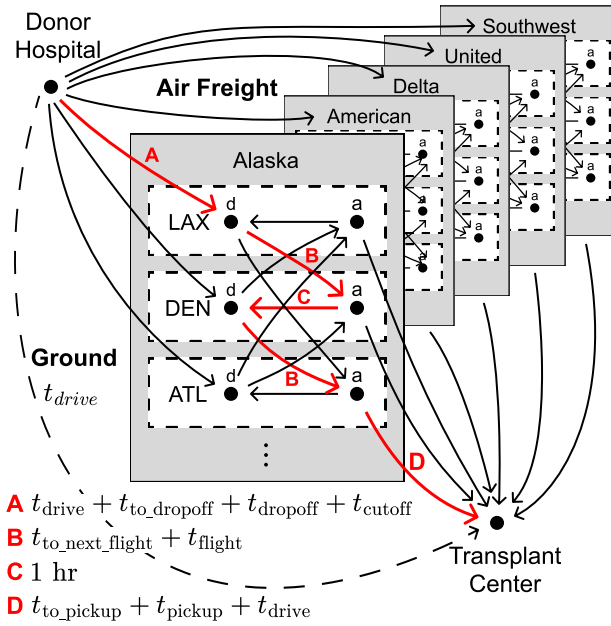


Figure 2 (Color online) Diagram of the Kidney Transportation Graph

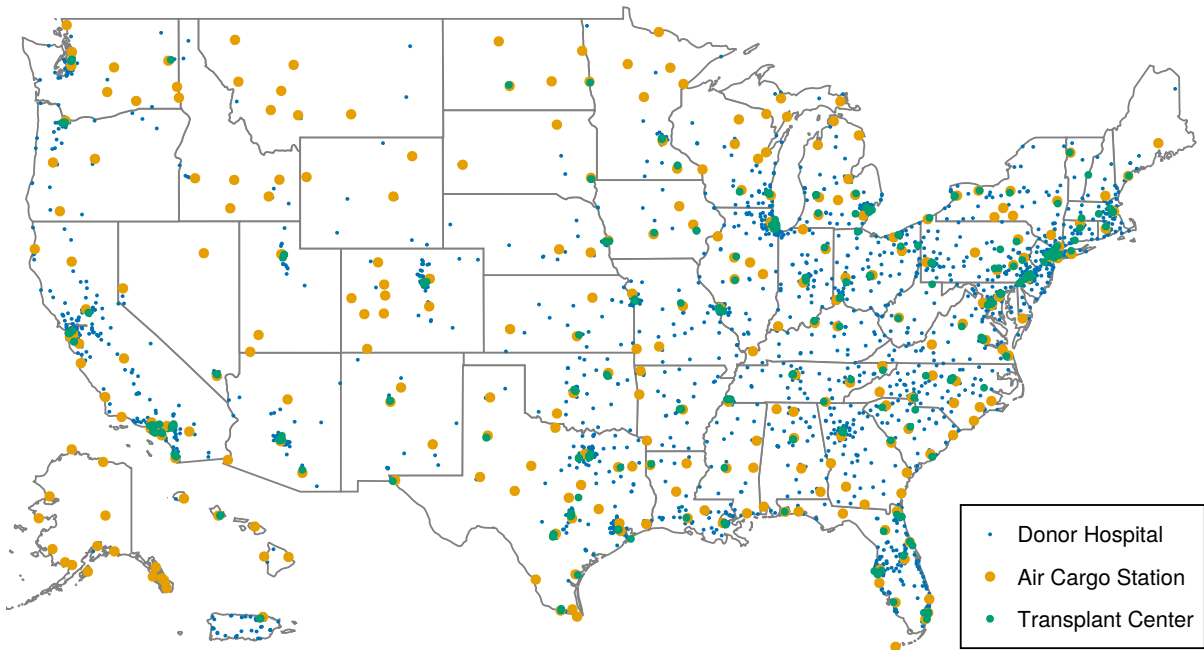


Figure 3 (Color online) Map of Donor Hospitals, Air Cargo Stations, and Transplant Centers in the United States

uses OpenStreetMap road network data, which was obtained for U.S. states and territories from Geofabrik (Geofabrik GmbH 2025).

Transportation by air freight involves several steps. An example air freight route is colored red in Figure 2. First, a courier drives the shipment to an airport and transfers it to an air freight

carrier. Five carriers are included in the model: Alaska Airlines, American Airlines, Delta Air Lines, United Airlines, and Southwest Airlines. Package drop-off occurs at the carrier’s dedicated cargo facility or, at smaller airports, their ticket counter. At each airport, carriers have different hours when drop-off may occur and cutoff times ahead of flight departure. Operational requirements of air cargo facilities was collected from the websites of the five carriers.

Five flight-departure airport vertices $v_{ap}^d \in V_{AP}$ are included in the graph for each airport, one for each air carrier. Edges connect from each donor hospital vertex $v_{dh} \in V_{DH}$ to each flight-departure airport vertex $v_{ap}^d \in V_{AP}$. The time required to transit from a donor hospital vertex $v_{dh} \in V_{DH}$ to a flight-departure airport vertex $v_{ap}^d \in V_{AP}$ is composed of four parts. First, donor hospital to airport drive time, $t_{drive}(v_{dh}, v_{ap}^d)$. Second, any time from arrival at the airport at time t until the package drop-off hours at the air cargo station of v_{ap}^d , $t_{to_dropoff}(t, v_{ap}^d)$. Third, the time required to perform the drop-off itself, $t_{dropoff}(v_{ap}^d)$. And finally, buffer time needed between drop-off and any flight departure, $t_{cutoff}(v_{ap}^d)$. Altogether, the weight for this kind of edge is $w_e(t) = t_{drive}(v_{dh}, v_{ap}^d) + t_{to_dropoff}(t + t_{drive}(v_{dh}, v_{ap}^d), v_{ap}^d) + t_{dropoff}(v_{ap}^d) + t_{cutoff}(v_{ap}^d)$.

After the shipment has been transferred to the air carrier, the package is transported in the cargo hold of commercial passenger flights. Edges connect from a carrier’s flight-departure airport vertices $v_{ap}^d \in V_{AP}$ to the same carrier’s flight-arrival airport vertices $v_{ap}^a \in V_{AP}$ with edge weights equal to $t_{to_next_flight}(t, v_{ap}^d, v_{ap}^a) + t_{flight}(t, v_{ap}^d, v_{ap}^a)$, the time until the carrier’s next available flight and the duration of the flight itself.

To calculate these edge weights, historical flight schedules from 2015 through 2024 for the five carriers were obtained from the Airline On-Time Performance database of the U.S. Bureau of Transportation Statistics (Bureau of Transportation Statistics 2025a,b). The database contains scheduled and actual departure and arrival times reported by air carriers that accounted for at least 1% of U.S. domestic scheduled passenger revenues prior to 2018 or 0.5% thereafter. The flight schedule for 2015 through 2017 came from the Reporting Carrier On-Time Performance table and the flight schedule for 2018 through 2024 came from the Marketing Carrier On-Time Performance table. The Marketing table explicitly identifies if a flight was operated on behalf of another carrier, and would therefore be available through their freight network. Flights of regional carriers from the 2015 to 2017 period, for which this information is not available, were associated with main carriers based on the rates in 2018 and later.

Flight transfers were assumed to require at least 1 hr. Hence, the graph contains edges with weight 1 from the flight-arrival vertex $v_{ap}^a \in V_{AP}$ to the flight-departure vertex $v_{ap}^d \in V_{AP}$ of each carrier at each airport. By transiting through one of these edges, the package may then transfer to a different flight through the flight-departure vertex.

The air freight process concludes with a courier retrieving the package at the destination airport and driving it to the recipient’s transplant center. As with the drop-off process, pick-up must occur within scheduled hours that vary by airport and carrier. The graph contains edges from all flight-arrival airport vertices $v_{\text{ap}}^{\text{a}} \in V_{\text{AP}}$ to all transplant center vertices $v_{\text{txc}} \in V_{\text{TXC}}$. The weight for an edge $e = (v_{\text{ap}}^{\text{a}}, v_{\text{txc}})$ is $w_e(t) = t_{\text{to_pickup}}(t, v_{\text{ap}}^{\text{a}}) + t_{\text{pickup}}(v_{\text{ap}}^{\text{a}}) + t_{\text{drive}}(v_{\text{ap}}^{\text{a}}, v_{\text{txc}})$.

To conclude this section, we note that separate flight-arrival and flight-departure vertices are included for the correctness of the routing algorithms. If a package arrives at an airport from a donor hospital, then the drop-off process must be performed. If a package arrived from another airport and is being transferred to a new flight, then only transfer time must elapse. Thus, the time needed for preflight activities at an airport depends on where the package arrived from, not just on when the package arrived. The routing algorithms require that a vertex’s outgoing edge weights depend only on time of arrival at that vertex, not the path taken to the vertex. This ensures alternative routes to the vertex can be compared only by how long it took to arrive at that vertex. The separate flight-arrival and flight-departure vertices allow transfer time to be included with edges between flight-arrival and flight-departure vertices and time associated with the drop-off process to be included in edges between donor hospitals and flight-departure vertices. The airport flight-departure vertex’s outgoing edges to other airports can then contain only the time until the next flight and flight duration, which do not depend on where a package arrived from.

Next, we discuss the algorithms that were used to compute fastest-paths through this graph.

3.2. Computational Implementation and Algorithms

Large-scale analyses of kidney allocation require planning millions of potential transportation routes, making computational speed critical. For example, in 2024 there were approximately 12,000 kidney donors with compatible candidates listed across an average of 130 transplant centers per donor. Assuming similar departure times for kidneys donated by the same donor, planning potential transportation routes over the year requires computing $12,000 \times 130 = 1.56$ million fastest paths.⁴

We use the Lazy A* algorithm (Cohen et al. 2014) to efficiently compute these routes. Lazy A* is particularly well suited to our setting because it avoids unnecessary queries to the historical flight schedule database. In a benchmark of 5,000 randomly selected routes, Lazy A* required an average of 3.7 ms per route and identified the fastest route in all cases. This was 47.2 times faster than Dijkstra’s algorithm and 7.9 times faster than standard A*. Additional details on the routing algorithm, implementation, and computational benchmarks are provided in section SM.1 of the electronic companion.

⁴ Because transportation time between a donor hospital and transplant center varies substantially depending on departure time, it is insufficient to consider only the product of the number of donor hospitals and transplant centers.

3.3. Model Validation

Validation of the transportation model was performed by comparing the durations of model-generated routes to historical transplant system logistics data. This data included the time that a kidney shipment was checked out at the origin, recorded by the OPTN barcode scanning system called TransNet, and the time that a kidney shipment was checked in at the recipient's transplant center, either recorded through TransNet or documented in the OPTN Transplant Recipient Registration form. We call the time that elapsed between these two events the kidney shipment's check-out time. Data for 846 shipments, stretching from September 16th, 2023 through December 31st, 2024 met inclusion criteria for validation (criteria below).

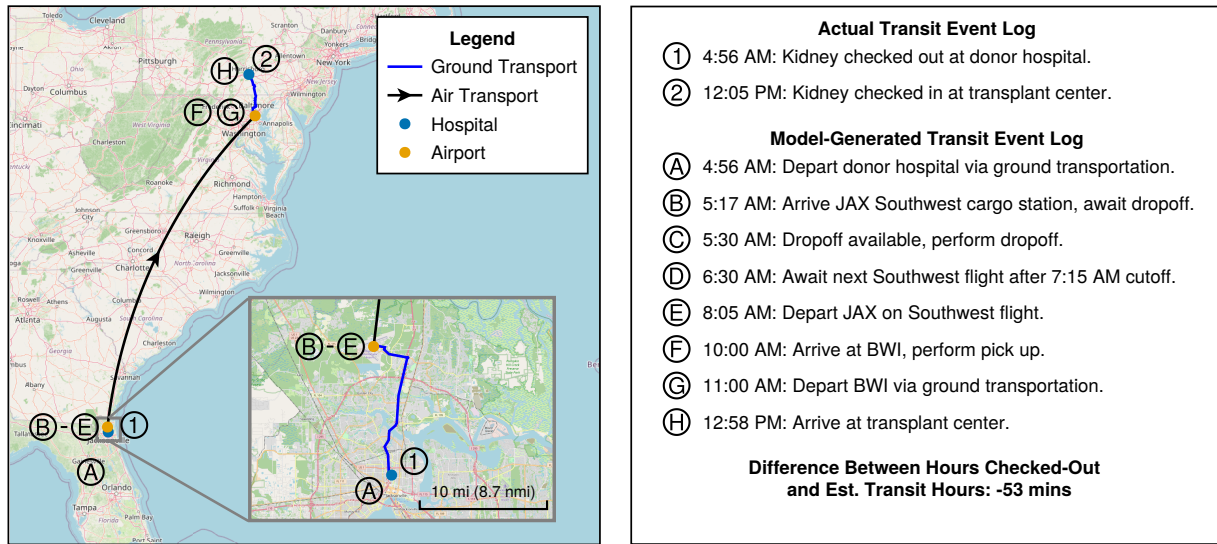


Figure 4 (Color online) Model-Generated Transportation Route for a Deceased-Donor Kidney

To illustrate this, Figure 4 shows a model-generated route for a kidney donated near Jacksonville, Florida in the fall of 2023. The historical data indicates the shipment was checked out at 4:56 AM and checked in at 12:05 PM later that day. A route was planned from the donor hospital to the transplant center departing at 4:56 AM. The route used Southwest air freight and entails first driving the shipment from the donor hospital to JAX airport, a wait for Southwest drop-off hours to begin, and then 1 hr for package drop-off. After a flight cutoff period ending at 7:15 AM, the route includes a flight bound for BWI at 8:05 AM and arrival at 10:00 AM. After 1 hr for shipment pick-up and a 1 hr 58 min drive, the route terminates at the transplant center. The historical data indicates the shipment check-out time was 7 hr 9 min, while the model estimates transport time is 8 hr 2 min. Hence, check-out time was 53 min shorter than estimated transportation time.

When interpreting these results, it is important to note limitations with the available data. First, the time from check-out to check-in does not precisely measure the time from departure

to arrival. For example, the time a shipment is checked in at the recipient's transplant center may be delayed many hours from the actual arrival time, for instance when a shipment arrives overnight. This means discrepancies between estimated transport time and recorded check-out time reflect measurement error as well as model error. GPS tracking data would provide a more reliable comparison, but we were unable to obtain this. Second, check-out time was only available for a relatively small fraction of kidney shipments. The sample of 846 kidney shipments represents 3% of the more than 27,000 kidney transplants performed during the period of the logistics data.

For inclusion, check-out time was required to be after the time the recipient was identified. Check-in time was required to be before the time the transplant was performed. TransNet check-out and check-in events were required to have donor and kidney identifiers that matched a historical kidney transplant. For TransNet check-in events, the receiver was also required to be the transplant center of the transplant recipient. Shipments were only included if arrivals were predicted to be between 5 AM and 6 PM in order to avoid delayed check-ins that may occur when a shipment arrives overnight. Finally, to filter outliers, we assumed no shipment could travel faster than 400 mph on average, or 80 mph for shorter travel distances (2-200 nmi) where ground transport would be used. When multiple check-out and check-in events were recorded per shipment only the latest check-out and earliest check-in were used.

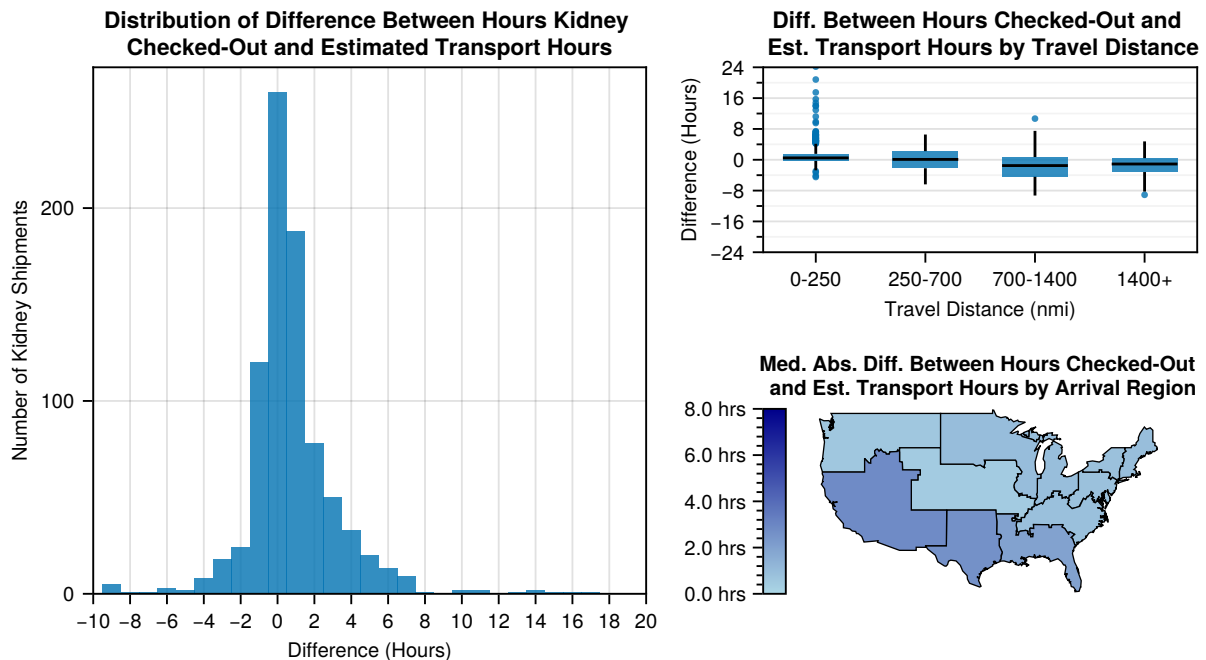


Figure 5 (Color online) **Transportation Model Validation Results**

Validation results are shown in Figure 5. The left plot shows the distribution of differences between historical check-out hours and estimated transportation times. The median error was 0.4 hr and the median absolute error (MedAE) was 0.9 hr. This suggests that check-out times tended to be slightly greater than estimated transportation times, but were generally similar to the check-out times observed historically. More sensitive to larger deviations, the mean absolute error (MAE) was 1.7 hr and the root mean squared error (RMSE) was 2.8 hr. Comparing two shipments, the shipment with longer estimated transportation time also had longer historical check-out time 78.2% of the time, or 85.0% when considering pairs with at least a one hour difference.

As transportation distance increased, the differences between estimated transportation time and historical check-out time grew moderately. The top right plot of Figure 5 shows box-and-whisker plots for the difference between historical check-out time and estimated transport time for four ranges of distance between donor hospital and transplant center. Between 0 and 250 nmi, MedAE was 0.8 hr, between 250 and 700 nmi it was 2.3 hr, between 700 and 1400 nmi it was 3.0 hr, and greater than 1400 nmi it was 2.9 hr. This indicates the model has reasonable accuracy across a broad range of distances and across both modes of transport.

Geographic variation in model accuracy was considered by calculating MedAE for shipments arriving in each OPTN region. This is depicted in bottom right plot of Figure 5. Of the 11 OPTN regions, 3 had a MedAE greater than 1 hr. These were the southwest (OPTN region 5), Texas-Oklahoma (OPTN region 4), and southern (OPTN region 3) regions. Collectively, the shipments arriving in these regions had a median travel distance of 734 nmi, so regional variation is explained by the length of the shipments recorded within each region.

These validation results indicate that, despite the inherent discrepancy between check-out time and true transport time, transportation times estimated by the model align reasonably well with historical shipment check-out times. For validation of the OSRM driving router in particular, we refer the reader to Fu et al. (2023). Finally we note that transplant system professionals have estimated that approximately 17–20% of deceased-donor kidneys were transported via air freight in 2018 and 2019 (Organ Transport Working Group 2025, Aleccia 2020). The transportation model predicts comparable values of 19% in 2018 and 18% in 2019. Together, these comparisons indicate that the model captures both the timing and modal characteristics of kidney transportation at a suitable level for our subsequent analyses. Further development may be desired for use in organ allocation policy or to support route planning in practice. More detailed validation may be possible in the future with GPS tracking data.

4. Optimizing Transplant System Logistics

In Section 4.1, we consider whether the geographic distribution of kidney transplants after the recent policy change that de-emphasized donor–recipient proximity could be achieved more efficiently. In Section 4.2, we consider more broadly whether a regionally balanced allocation could be logistically feasible.

4.1. Efficiently Achieving Effects of Current Policy

In March 2021, the OPTN implemented a new kidney allocation policy that reduced the importance of transplant candidate proximity to the donor. Through broader distribution of kidneys, policy-makers sought to provide more consistent access to kidney transplantation regardless of where a patient lived. In this section, we consider whether the patients who received transplants under this broader distribution could have been served in a more efficient manner.

Specifically, we examine whether the kidneys transplanted in 2024 could have been reassigned among the recipients of those kidneys to reduce transportation complexity, without meaningfully extending recipients’ waiting times or lowering the quality of kidney they received. In other words, we consider whether the effects of the broader distribution could be achieved without actually broadly distributing kidneys. Improved logistics could potentially reduce delayed graft function, reduce kidney non-use, and decrease out-of-sequence allocation. We first formulate this as an optimization problem and then present results.

4.1.1. Problem Formulation We formulate this as a bipartite assignment problem between donated kidneys and transplant recipients:

$$\min_x \sum_{i \in I} \sum_{j \in I} f_k(t_{ij}) x_{ij} \quad (1)$$

$$\text{s.t.} \quad \sum_{i \in I} x_{ij} = 1 \quad \forall j \in I \quad (2)$$

$$\sum_{j \in I} x_{ij} = 1 \quad \forall i \in I \quad (3)$$

$$x_{ij} \leq a_{ij} \quad \forall i \in I, \forall j \in I \quad (4)$$

$$\sum_{i \in I} \sum_{j \in I} f_{ij} x_{ij} \leq F \quad (5)$$

$$x_{ij} \in \{0, 1\} \quad \forall i \in I, \forall j \in I. \quad (6)$$

Here, I denotes the set of deceased-donor kidney transplants during 2024 and the decision variables are:

$$x_{ij} = \begin{cases} 1 & \text{if the kidney of transplant } j \text{ is assigned to the recipient of transplant } i, \\ 0 & \text{otherwise.} \end{cases}$$

The problem parameters are:

- $t_{ij} \geq 0$: estimated transportation time if kidney j is assigned to recipient i ,
- $a_{ij} \in \{0, 1\}$: feasibility indicator equal to 1 if eligibility criteria are met, the recipient would wait at most 8 additional weeks, and donor KDPI would increase by at most 15 percentiles (see ,
- $f_{ij} \in \{0, 1\}$: indicator equal to 1 if air freight is used under the predicted route,
- $F \in \mathbb{Z}$: estimated number of historical shipments that used air freight in 2024.

Constraint (2) ensures that each kidney is assigned to exactly one recipient. Constraint (3) ensures that each recipient receives exactly one kidney. Constraint (4) enforces clinical feasibility, as well as limits on additional waiting time and decrease in organ quality. Constraint (5) ensures that estimated air freight usage does not exceed historical levels. Three transportation time cost functions are considered to either minimize total transportation time ($f_1(t) = t$), penalize longer shipments ($f_2(t) = \min\left(\frac{e^{0.3t}-1}{0.3}, 3000\right)$), or favor shorter shipments ($f_3(t) = \frac{e^{-0.3t}-1}{-0.3}$). These functions are plotted in Figure 6.

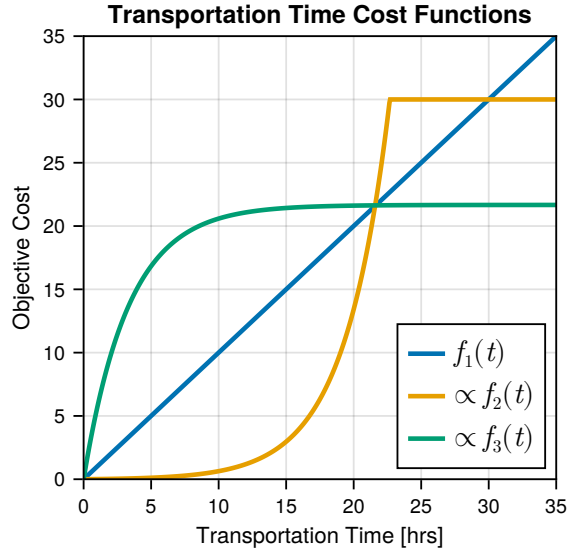


Figure 6 (Color online) Transportation Time Cost Functions

4.1.2. Results The integer optimization problems with transportation time cost functions f_1 , f_2 , and f_3 were solved using the JuMP optimization package for the Julia programming language with the Gurobi optimization solver (Lubin et al. 2023, Bezanson et al. 2017, Gurobi Optimization, LLC 2026). Results are shown in Figure 7. The kidney shipments performed in 2024 are shown in the map in Panel A, with edges connecting the donor hospital and recipient transplant center for each historical transplant performed that year. Panel B shows the kidney shipments necessary under the f_1 -optimal reassignment. Total transportation time is reduced from 131,000 hr to just over 89,000 hr (-31.7%).

The distribution of estimated transportation times under the historical assignment, as well as under the f_1 -optimal (Min Total, T), f_2 -optimal (Min Long, L), and f_3 -optimal (Max Short, S) reassignments, are shown in Panel C. Shorter shipments become substantially more frequent. The number of shipments estimated to take less than 1 hr increases from 4,704 to 9,529 (+102.6%) under the f_1 -optimal reassignment, and to 10,468 (+122.5%) under the f_3 -optimal reassignment. Meanwhile, the number of shipments estimated to take longer than 10 hr, as well as the number longer than 12 hr, decreases significantly. Under the f_1 -optimal reassignment, the number of shipments with estimated transport times longer than 10 hr is reduced from 2,065 to 688 (-66.7%), while the number longer than 12 hr is reduced from 1,307 to 376 (-71.2%). This reduction is even more pronounced under the f_2 -optimal reassignment. The number of shipments exceeding 10 hr is reduced from 2,065 to 446 (-78.4%), and the number of shipments exceeding 12 hr is reduced from 1,307 to 232 (-82.2%). Reductions in very long shipments may be particularly important because prolonged transportation times can substantially increase cold ischemic time and introduce greater logistical uncertainty.

In Panel D, box-and-whisker plots of the great-circle distance between donor hospitals and recipient transplant centers are shown under the historical and optimized reassignments. Distances between donor hospitals and recipient transplant centers decrease substantially under the optimized assignments. For historical deceased-donor kidney transplants, the median distance is 123 nmi. This is reduced to 44 nmi (-64.2%) under the f_1 -optimal assignment, to 63 nmi (-48.8%) under the f_2 -optimal assignment, and to 31 nmi (-74.8%) under the f_3 -optimal assignment. These findings indicate that the transportation time reductions are not achieved solely by transporting kidneys more efficiently along existing shipment routes, but also by substantially reducing the geographic distance between donors and recipients.

Reductions in air freight usage are also observed under the optimized assignments. Panel E shows the number of shipments under each assignment whose model-generated transportation route used air freight. Under the historical assignment, 3,325 shipments are estimated to use air freight. This is reduced to 2,031 (-38.9%), 2,171 (-34.7%), and 2,404 (-27.7%) under the f_1 -optimal, f_2 -optimal, and f_3 -optimal assignments, respectively. If air freight usage were explicitly included in the optimization objective, additional reductions might be possible. Reductions in air freight usage may substantially improve the reliability of kidney transportation logistics, as air freight is a major source of transportation failures.

These transportation reductions are achieved while providing the same patients with kidney transplants similar to those they received historically. The optimization problem enforced that for each recipient, donor KDPI would not increase by more than 15 percentiles relative to the historical transplant. Panel F shows a heatmap comparing recipients' historical donor KDPI with donor

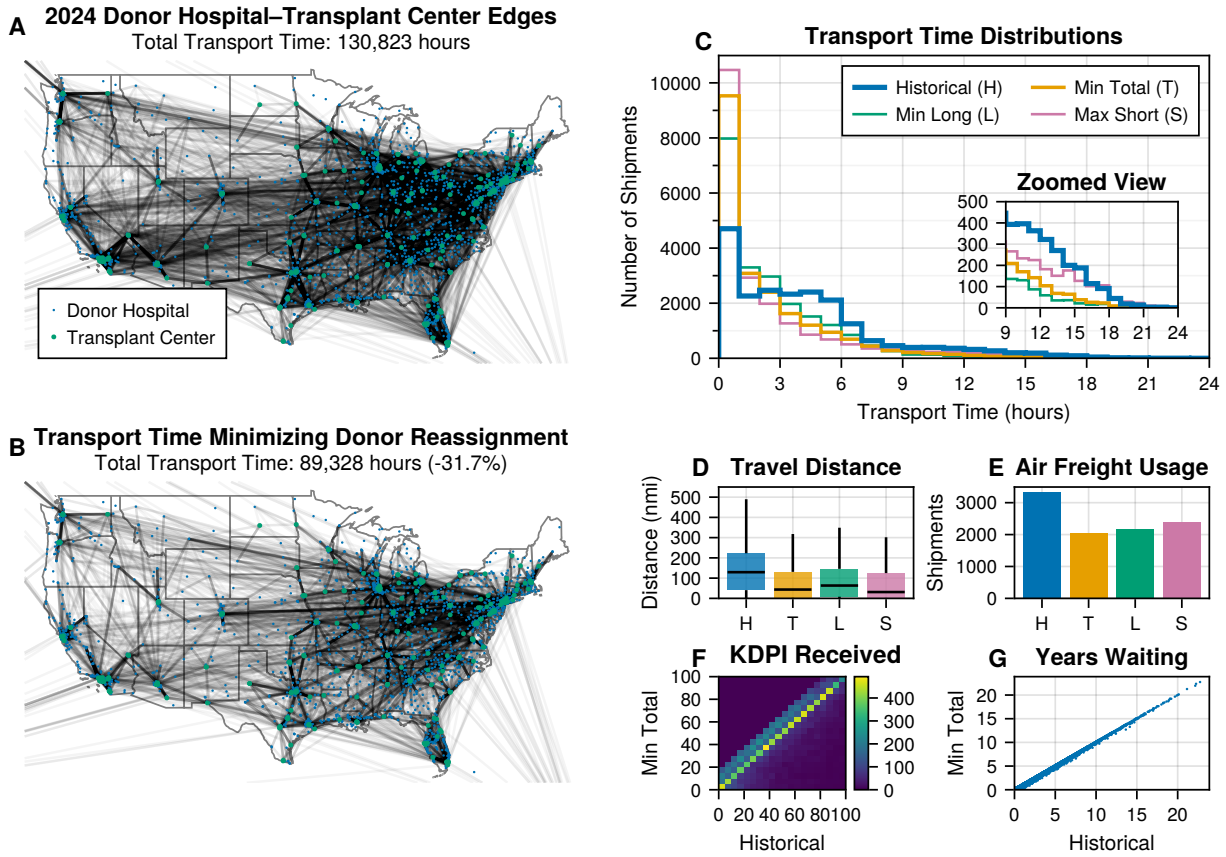


Figure 7 (Color online) Comparison of the 2024 Donor–Recipient Assignment with Three Optimized Assignments

KDPI under the f_1 -optimal reassignment. Under the f_1 -optimal reassignment, 42.8% of recipients receive a kidney from a donor with a higher KDPI than they received historically, with a median increase of 9 percentiles among those recipients.

Additionally, each recipient’s waiting time was constrained not to increase by more than 8 weeks, a small fraction of typical waiting times. Panel G compares recipients’ historical waiting time at transplant with their waiting time under the f_1 -optimal reassignment. Among the 44.9% of recipients whose waiting time increases, the median increase in waiting time is approximately 30 days. Relative to total waiting time at transplant, this amounts to an average increase of 1.3% among recipients with increased waiting time.

These results demonstrate that substantial improvements in logistical efficiency may have been possible while providing transplants to the same patients that received transplants under broader distribution. This suggests that some of the logistical burden associated with broader kidney distribution may not be necessary to achieve similar equity benefits.

At the same time, geographic access to kidney transplantation remains uneven under current policy. A natural question is whether further improvements in geographic equity would require substantially greater transportation demands. In the next section, we evaluate the logistical implications of pursuing a more geographically balanced distribution.

4.1.3. Sensitivity Analysis In the main analysis, we require a kidney transplant recipient to wait at most 8 additional weeks and the donor KDPI of their received kidney to increase by at most 15 percentiles. To test sensitivity of results to these values, we solve the assignment problem with objective function f_1 using alternative values of 4, 6, 8, 10, and 12 weeks and 5, 10, 15, 20, and 25 KDPI percentiles. Total transportation time varies from 104,176 hr (-20.5%) with the most restrictive requirements to 85,595 hr (-34.7%) with the least restrictive. The number of shipments exceeding 10 hr varies from 929 (-55.0%) to 599 (-71.0%). Tables SM.2–SM.9 in the electronic companion provide additional results.

4.1.4. Robustness Analysis This model does not capture that transplant offers are often declined and transplant candidates must be prioritized without precise knowledge of future donations. We assess the robustness of these results to substantial perturbations in donor–recipient assignments to determine whether these outcomes may be achievable in practice. A sequence of $k = 1, \dots, 10$ assignment problems are solved, resulting in optimal assignments $x^1, \dots, x^{10}$. Assignment problem k follows the formulation given above, except with the following added constraints:

$$\sum_{(i,j):x_{i,j}^l=1} (1 - x_{ij}) \geq 18,000 \quad \forall l = 1, \dots, k - 1.$$

These constraints require that at least 18,000 (85.3%) of each of the prior optimal assignments are not used. From $k = 1$ to $k = 10$, we find that total transportation time increases from 89,329 hr (-31.7%) to 97,077 hr (-25.8%) and that the number of shipments that exceed 10 hr increases from 688 (-66.7%) to 771 (-62.7%). Figure SM.1 in the electronic companion provides additional results.

4.2. Feasibility of a Regionally-Balanced Kidney Allocation

Despite the 2021 policy change allowing for broader geographic distribution, access to kidney transplantation has remained uneven. In 2024, regional transplant counts ranged from nearly 37.4% below to 59.1% above expected levels given the candidates listed in each region. Here we investigate whether a more regionally balanced distribution can be achieved without imposing impractical logistical requirements.

We consider reassigning kidneys transplanted in 2024 among candidates listed that year such that (1) total transportation time is minimized, (2) donor–recipient pairings remain similar at a

national level, with respect to organ quality and recipient prioritization factors, to those observed in 2024, and (3) transplants are distributed geographically in proportion to the candidates listed in each region. This way the resulting allocation remains as consistent as possible with current policy, apart from achieving a more balanced geographic distribution. We next define this balanced distribution precisely, formulate the associated optimization problem, and assess the resulting assignment.

4.2.1. Defining A Regionally Balanced Distribution Regional transplant targets were calculated across different transplant categories in a manner that maintained the donor–recipient characteristics of the 2024 transplants at a national level, while redistributing transplants proportionally to the candidates listed in each region. Targets were updated monthly, based on the candidates listed at the beginning of each month, to account for waitlist dynamics. Table 1 illustrates how these targets were defined for one transplant category (for simplicity, annual targets are shown rather than monthly).

Region	Candidates	Share of Candidates	Transplants	Share of Transplants	Expected Transplants
1	73	3.6%	4	0.9%	16.6
2	249	12.2%	45	9.7%	56.8
3	116	5.7%	53	11.4%	26.4
4	140	6.8%	56	12.0%	31.9
5	873	42.7%	110	23.6%	199.0
6	24	1.2%	17	3.6%	5.5
7	119	5.8%	44	9.4%	27.1
8	33	1.6%	17	3.6%	7.5
9	189	9.2%	31	6.7%	43.1
10	84	4.1%	27	5.8%	19.2
11	144	7.0%	62	13.3%	32.8
U.S. Total	2,044	100%	466	100%	466

Table 1 Regional Transplant Target Determination for a Kidney Transplant Category

Transplant targets were calculated for transplant categories that were based on kidney quality and recipient priority factors. Kidney quality was defined by donor KDPI grouped into four bins (0-20, 21-50, 51-80, and 81-100) and recipient priority factors were those used for ranking under the current kidney allocation policy, listed in Table 2. For example, the transplant category of Table 1 requires donor KDPI between 21 and 50 and recipients who are adults with blood type O, have waited at least 8 years, have cPRA less than or equal to 20%, and do not meet EPTS Top 20, Post-Transplant Safety Net, or Prior Organ Donor requirements. The regional transplant target for category $n \in N$, month $m \in \{1, \dots, 12\}$, and each OPTN-defined region $r \in \{1, \dots, 11\}$ will be denoted $V_{n,m}^r$.

A regional target $V_{n,m}^r$ was calculated by first determining the number of transplants V_n that occurred within category n in 2024, as well as the number of candidates $c_{n,m}^r$ waiting at the

	Transplant Recipient Characteristics
1	Age (Pediatric or Adult)
2	Waiting Time at Transplant (0-2, 2-4, 4-6, 6-8, or 8+ yr)
3	cPRA (0-20%, 21-79%, 80-97%, 98%, 99%, or 100%)
4	ABO (A, B, AB, or O)
5	EPTS Top 20 (yes/no)
6	Post-Transplant Safety Net (yes/no)
7	Prior Organ Donor (yes/no)

Table 2 Characteristics of Kidney Candidates Used for Transplant Categorization

beginning of month m in region r that meet the criteria for category n . For Table 1, the total number of transplants performed in 2024 within the relevant category is listed under the Transplant column in the U.S. Total row and was $V_n = 466$. The number of candidates listed at the beginning of January 2024 in each region that meet the table’s category are shown in the Candidates column. A transplant target was calculated with the formula:

$$V_{n,m}^r = \frac{V_n}{12} \frac{c_{n,m}^r}{\sum_{r=1}^{11} c_{n,m}^r} \quad \forall n \in N, m \in \{1, \dots, 12\}, r \in \{1, \dots, 11\}.$$

For Table 2, annual regional targets computed with this formula (except with no division by 12) are found in the column Expected Transplants.

This approach preserves the total number of transplants within each category nationally, while allocating transplants across regions in proportion to the share of relevant candidates listed in each region.

4.2.2. Problem Formulation The problem was formulated as a sequence of 12 optimal assignment problems, one for each month of 2024. In each month, the objective was to minimize total transportation time and deviation from regional transplant targets while respecting OPTN match eligibility requirements. Let J_m denote the set of deceased-donor kidneys transplanted in month m . Let K be the set of patients with at least one kidney registration during 2024, I the set of registrations (one patient may register at multiple transplant centers), and I_k be the set of registrations corresponding to patient $k \in K$.

Whereas regional transplant targets were computed using joint categorization by KDPI and 7 recipient characteristics, yielding 7,680 distinct transplant types, the optimization formulation marginalized over recipient characteristics. That is, a single transplant could satisfy the criteria for multiple marginal categories. For example, a transplant involving a 0-20 KDPI kidney allocated to a pediatric candidate with 0-20% cPRA would satisfy both the category for 0-20 KDPI transplants to pediatric candidates and the category for 0-20 KDPI transplants to candidates with 0-20% cPRA. As a result, each transplant belonged to 7 of the 80 possible marginal categories denoted by S .

Let $G_{s,m}^r \subset I \times J_m$ denote the set of registration-donor pairings in region r during month m that meet the criteria for marginalized category $s \in \{1, \dots, 80\}$. Let $N_s \subset N$ denote the set of joint transplant categories consistent with marginalized category s , and define $V_{s,m}^r = \sum_{n \in N_s} V_{n,m}^r$. These are the marginalized regional transplant targets used in the optimization problem.

The decision variables of the assignment problem for month m were:

$$x_{ij} = \begin{cases} 1 & \text{if the kidney of transplant } j \in J_m \text{ is assigned to the patient with registration } i \in I, \\ 0 & \text{otherwise.} \end{cases}$$

The corresponding problem parameters were:

- $t_{ij} \geq 0$: estimated transportation time if kidney j is assigned to candidate i ,
- $a_{ij} \in \{0, 1\}$: feasibility indicator equal to 1 if registration i satisfies eligibility criteria for kidney j and was not assigned a transplant in an optimized assignment for an earlier month,
- $V_{s,m}^r \geq 0$: target number of type s transplants during month m in region r , updated based on optimized assignments from earlier months and newly listed candidates,
- $\alpha > 0$: weighting parameter controlling the relative importance of minimizing transportation time and matching regional transplant targets.

The assignment problem for month m was formulated as the following integer optimization problem:

$$\min_x \quad \alpha \sum_{i \in I} \sum_{j \in J_m} t_{ij} x_{ij} + \sum_{s \in S} \sum_{r=1}^{11} \left| \sum_{(i,j) \in G_{s,m}^r} x_{ij} - V_{s,m}^r \right| \quad (7)$$

$$\text{s.t.} \quad \sum_{i \in I} x_{ij} = 1 \quad \forall j \in J_m \quad (8)$$

$$\sum_{i \in I_k} \sum_{j \in J_m} x_{ij} \leq 1 \quad \forall k \in K \quad (9)$$

$$x_{ij} \leq a_{ij} \quad \forall i \in I, \forall j \in J \quad (10)$$

$$x_{ij} \in \{0, 1\} \quad \forall i \in I, \forall j \in J. \quad (11)$$

The objective trades off total transportation time against deviations from regional transplant targets across marginal transplant categories. Constraint (8) ensures that each kidney transplanted in month m is assigned to exactly one registration. Constraint (9) ensures that each patient receives at most one transplant in month m . Constraint (10) enforces clinical feasibility and prevents registrations assigned a transplant in an earlier month from being assigned again.

4.2.3. Results The integer optimization problems were solved using the JuMP optimization package for the Julia programming language with the Gurobi optimization solver on a node from MIT Lincoln Lab's SuperCloud computing cluster (Lubin et al. 2023, Bezanson et al. 2017, Gurobi Optimization, LLC 2026, Reuther et al. 2018). Results are shown in Figure 8.

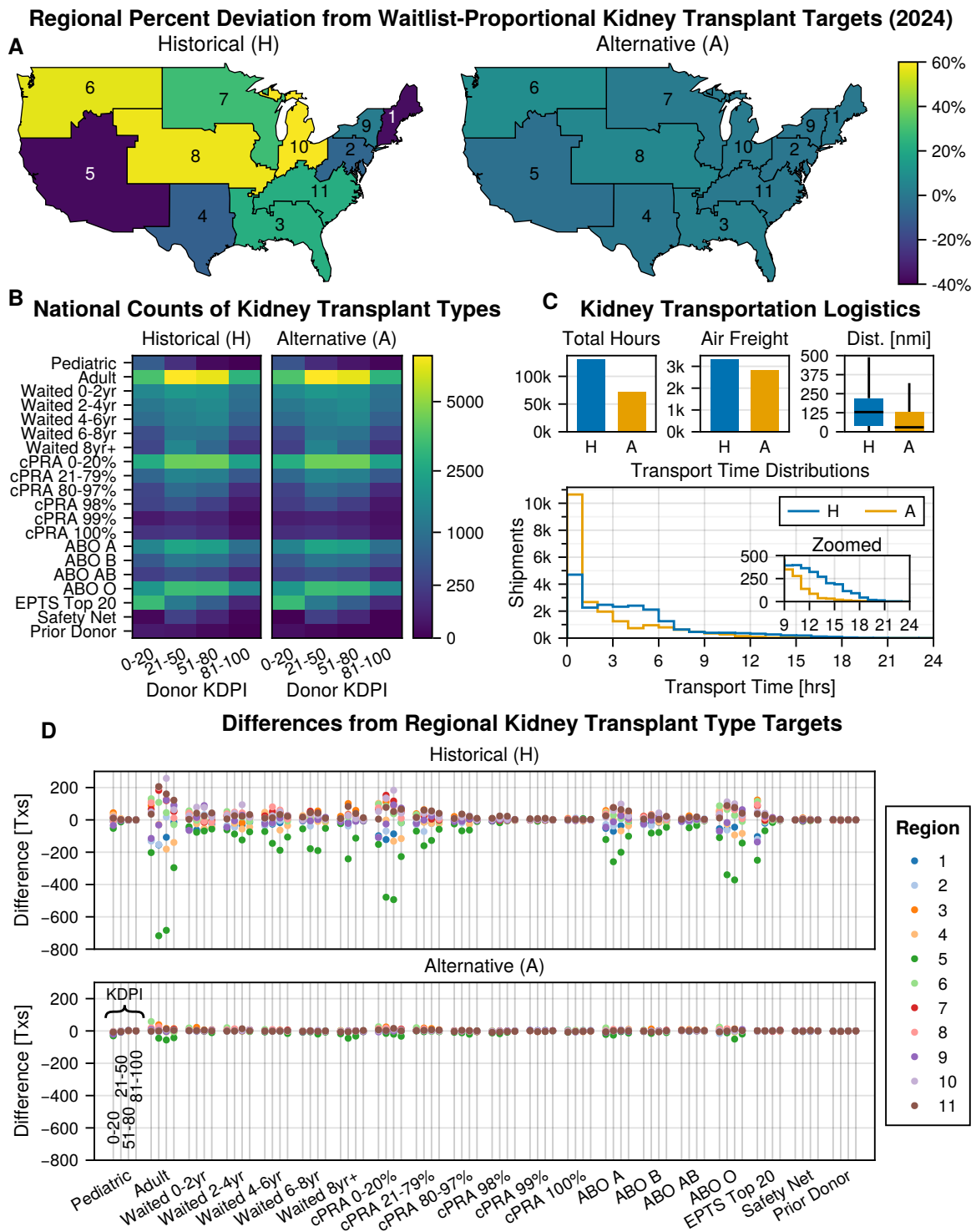


Figure 8 (Color online) Comparison of the 2024 Donor–Recipient Assignment with a Regionally Balanced Assignment

First, the number of transplants received in each region is substantially closer to expected levels under the optimized assignment. The maps in Panel A compare the number of transplants received in each region with the sum of monthly transplant targets across all transplant categories in that region. Under the historical assignment, region 5 experiences the largest deficit. The sum of transplant targets in region 5 is 5,240, but only 3,280 kidney transplants are performed, corresponding to a deficit of -37.4% . Under the optimized assignment, the largest deficit also occurs in region 5, but is substantially reduced. The sum of transplant targets is 4,973, while 4,800 transplants are performed, corresponding to a deficit of -3.5% .

Under the historical assignment, the largest excess occurs in region 10. Patients listed in region 10 receive 1,598 kidney transplants, while the sum of transplant targets over the year is only 1,004, corresponding to an excess of $+59.2\%$. Under the optimized assignment, the largest excess occurs in region 3. There, the sum of transplant targets is approximately 2,397, while about 2,474 occurs, an excess of roughly $+3.1\%$.

While transplants are distributed more proportionally to the candidates listed in each region, the types of transplants performed nationwide remain largely unchanged. This is illustrated in Panel B. The colors in the heatmap indicate the number of transplants of each category that occur historically or under the optimized assignment. The vertical axis shows recipient characteristics, while the horizontal axis shows kidney quality as measured by donor KDPI. The optimized assignment produces very similar numbers of transplants across both smaller, high-priority transplant types (such as those of pediatric recipients receiving high-quality 0–20 KDPI kidneys) and more common transplant types (such as adult recipients receiving a 21–50 KDPI kidney). This demonstrates that the optimized assignment preserves the overall composition of transplant types observed historically.

The transportation requirements of the optimized assignment are no greater than those of the historical assignment. In fact, transportation requirements are substantially reduced, as shown in Panel C. Total transportation time is reduced from 130,823 hr to 71,755 hr (-45.2%), the number of air freight shipments is reduced from 3,325 to 2,813 (-15.4%), and the median transportation distance is decreased from 129 nmi to 30 nmi (-76.7%). Additionally, the number of shipments estimated to require more than 10 hours of transportation is reduced from 2,065 to 642 (-68.9%), while the number estimated to require more than 12 hours is reduced from 1,307 to 228 (-82.6%).

Panel D provides a more detailed view of the geographic distribution of transplants. The plot displays the difference between regional transplant target and the number of transplants received in each region for the transplant categories in Panel B. Under the optimized assignment, points fall much closer to a horizontal line at 0, indicating a substantially more proportional geographic distribution of transplants than under the historical assignment.

5. Conclusion

Seeking to address geographic disparities in access to transplantation, recent reforms to U.S. kidney allocation policy have reduced the importance of geographic proximity between transplant candidates and donors. This shift has challenged the time sensitive and high stakes logistics of the U.S. transplant system. In this paper, we consider whether geographic disparities could be addressed more efficiently. Our results suggest that they can. We find that both the geographic distribution of transplants under current policy and a distribution with waitlist-proportional regional transplant counts could be achieved with substantially less transportation than required under current policy.

These analyses are made possible by a national-scale model of deceased-donor kidney transportation that integrates U.S. road networks, historical flight schedules, and air cargo station operations. Using the Lazy A* routing algorithm, the model evaluates transportation routes in milliseconds, making it feasible to account for transportation time across millions of potential donor-recipient assignments. This model provides a new tool for evaluating the logistical implications of kidney allocation policies.

Our analysis has two important limitations. First, the optimized allocations are retrospective, so they rely on information about future organ supply that would not be available in practice. Second, kidney allocation involves decentralized decision making, with transplant centers independently choosing to accept organ offers. Nevertheless, robustness analyses indicates that substantial efficiency gains persist even after tens of thousands of optimal assignments are eliminated. This suggests that the qualitative conclusions are likely robust to both online and decentralized decision-making. These findings motivate future work to translate these gains into practical allocation policies.

Supplemental Materials

SM.1. Routing Algorithms

The transportation model was implemented in the Julia programming language (Bezanson et al. 2017). The transportation graph has a regular structure that allows neighbors to be generated on demand, so it was represented implicitly via successor functions rather than stored explicitly, for example as an adjacency matrix. Drive times between donor hospitals, airports, and transplant centers were precomputed with OSRM and stored in a matrix. Cargo station operational requirements (weekly drop-off, acceptance, cutoff, and pick-up times) for each air freight carrier were stored within airport objects. The flight schedule for the five air carriers from 2015 through 2024 was stored in a SQLite database so that the full schedule, more than 10 GB in total, would not need to be loaded into memory. The database was indexed by origin airport, destination airport, carrier, and departure timestamp to enable fast retrieval of the next outbound flight during node expansion.

During model development, four algorithms were used for computing fastest-paths: Dijkstra’s algorithm, A*, Lazy A*, and Weighted Lazy A* (Dijkstra 1959, Hart et al. 1968, Cohen et al. 2014). Ultimately, Lazy A* was used in practice. To explain the benefits of Lazy A*, we review the classical Dijkstra’s algorithm and A*.

To compute a path from an origin v_o to a destination v_d , Dijkstra’s algorithm and A* operate similarly. Beginning from the origin vertex v_o , a sequence of nodes in the graph are visited while a prioritized list of the vertices to visit subsequently, often called the frontier, is maintained. When a vertex v is visited, that vertex is expanded, meaning its neighbors $N(v)$ are either added to the frontier or, if already added, their priority in the frontier may be updated. The priority of a neighbor $v_n \in N(v)$ already in the frontier is updated if traveling through v , the vertex currently being visited, offers a better path to v_n than passing through any previously visited vertices.

The algorithms differ in how vertices in the frontier are prioritized for being visited. With Dijkstra’s algorithm, a vertex v in the frontier is prioritized according to the time it would take to reach that vertex v from the origin v_o (if only passing through the vertices visited so far) which will be denoted $t(v_o, v)$. This results in vertices being visited in a fashion that radiates out from the origin vertex v_o in all directions, regardless of the desired destination. The A* algorithm takes into account the destination v_d to direct the sequence towards the destination, reducing the number of visited vertices. A* prioritizes a vertex v in the frontier by an optimistic estimate (a lower bound) of the time to travel from the origin vertex v_o to the destination vertex v_d if routed through vertex v , $\underline{t}(v_o, v_d; v)$. This lower bound is determined by adding a lower bound on the time to travel from

vertex v to the destination vertex v_d , which we denote $\underline{t}(v, v_d)$, to the quantity used in Dijkstra's algorithm $t(v_o, v)$, the best travel time to vertex v from origin v_o identified so far.

The lower bound $\underline{t}(v, v_d)$ used in A* is called the heuristic. For the kidney transportation model, this heuristic was computed using Dijkstra's algorithm to find fastest-paths from each airport vertex $v_{ap} \in V_{AP}$ to each transplant center vertex $v_{txc} \in V_{TXC}$ in a transportation graph modified so that optimistic lower bounds were produced. Specifically, air carrier drop-off, pick-up, and flights were assumed to be immediately available and flights were assumed to take an optimistically small amount of time compared to the historical flight schedule. Optimistic flight times were calculated as the larger of the minimum observed flight duration or 25% less than the average flight duration between two airports. While 99.99% of flights were measured to take longer than these optimistic flight times, in rare cases it is possible for the heuristic to overestimate the time to reach the destination. This means that routes found using the heuristic may not be the fastest path possible.

While the A* algorithm reduces the number of vertices visited, a substantial fraction of computation time in this application is spent querying the flight schedule database unnecessarily. When expanding a vertex v under Dijkstra's algorithm or A*, the frontier priority score for each neighbor $v_n \in N(v)$ must be computed. For Dijkstra's algorithm this is $t(v_o, v) + t(v, v_n)$, the time from the origin vertex v_o to the vertex being visited v plus the time from the currently visited vertex v to the neighbor v_n . For A*, the frontier priority score is $t(v_o, v) + t(v, v_n) + \underline{t}(v_n, v_d)$, the same as Dijkstra's with the addition of a lower bound on time from the neighbor v_n to the destination v_d . While $t(v_o, v)$ is stored from earlier (when vertex v was added to the frontier and potentially as an update when better paths to v were identified) and $\underline{t}(v_n, v_d)$ is a simple lookup in a heuristic matrix, the value $t(v, v_n)$ must be computed at the time v is expanded. For expansion of a flight-departure vertex $v_{ap}^d \in V_{AP}$, the times to all other airports needs to be determined which leads to many unnecessary flight database queries.

Lazy A* and Weighted Lazy A* reduces the number of flight database queries that must be performed by breaking the visiting process into two stages. Like A*, Lazy A* prioritizes a vertex v in the frontier according to $\underline{t}(v_o, v_d; v)$, an optimistic estimate of travel time from origin v_o to destination v_d through v . However, when expanding a vertex v , the priority of a neighbor $v_n \in N(v)$ is determined using $t(v_o, v) + \underline{t}(v, v_n) + \underline{t}(v_n, v_d)$, the sum of time from the origin vertex v_o to the vertex being visited, the heuristic from neighbor to the destination $\underline{t}(v_n, v_d)$, and, differing from A*, $\underline{t}(v, v_n)$ an optimistic lower bound on travel time from the vertex being visited v to the neighbor v_n rather than the actual vertex-to-neighbor travel time. A flight schedule database lookup is avoided by using the same optimistic flight times that were used when computing the A* heuristic in place of a database query to compute actual travel times between two airports.

Ultimately, the estimate $\underline{t}(v, v_n)$ is replaced with the true value $t(v, v_n)$ but only if that airport edge (v, v_n) appears promising. If v_n rises to the top of the frontier and is therefore visited, it is not yet expanded. Instead the flight database is queried, the true value $t(v, v_n)$ is calculated, and the priority of v_n in the frontier is updated. Only if that neighbor is visited a second time is it actually expanded, with its neighbors $N(v_n)$ added to the frontier. This two stage approach introduces additional complications, for instance multiple copies of a vertex must be tracked. The interested reader is referred to Cohen et al. (2014). Altogether, this significantly reduces the number of flight database queries that are needed and accelerates routing.

Weighted Lazy A* is a greedy variant of Lazy A* akin to Weighted A*. It emphasizes finding reasonably fast routes quickly over finding optimal routes more slowly. It does this by altering the frontier prioritization score for a vertex v to $t(v_o, v) + \lambda \underline{t}(v, v_d)$, applying a weight $\lambda > 1$ to the time from the vertex v to the destination v_d . In essence, it considers the remaining time to the destination v_d to be more important than time from origin v_o to the vertex v .

Algorithm	Percent Routes Optimal	Optimality Gap (relative)		
		Mean	Median	Max
Dijkstra	100%	0 min (0%)	0 min (0%)	0 min (0%)
A*	100%	0 min (0%)	0 min (0%)	0 min (0%)
Lazy A*	100%	0 min (0%)	0 min (0%)	0 min (0%)
Lazy Weighted A* ($\lambda = 1.1$)	97.5%	0.17 min (0.02%)	0 min (0%)	26 min (4.00%)
Lazy Weighted A* ($\lambda = 1.25$)	92.1%	1.29 min (0.18%)	0 min (0%)	187 min (39.92%)
Lazy Weighted A* ($\lambda = 1.5$)	80.6%	6.00 min (0.82%)	0 min (0%)	300 min (39.92%)
Lazy Weighted A* ($\lambda = 2$)	64.4%	22.79 min (0.03%)	0 min (0%)	350 min (50.44%)

Algorithm	Run Time			Nodes Expanded			Flight Database Queries		
	Mean	Median	Max	Mean	Median	Max	Mean	Median	Max
Dijkstra	174.8ms	148.0ms	803.0ms	347.5	283	1,688	50,250	42,279	219,711
A*	29.4ms	19.5ms	620.3ms	37.2	22	1,285	8,926	5,842	240,342
Lazy A*	3.7ms	2.1ms	99.3ms	36.2	21	1,284	268	87	9,881
Lazy Weighted A* ($\lambda = 1.1$)	3.2ms	1.7ms	97.5ms	29.8	16	1,282	219	61	9,331
Lazy Weighted A* ($\lambda = 1.25$)	2.6ms	1.4ms	84.7ms	23.1	12	1,242	167	39	8,031
Lazy Weighted A* ($\lambda = 1.5$)	2.0ms	1.1ms	64.9ms	16.0	9	1,160	112	22	5,242
Lazy Weighted A* ($\lambda = 2$)	1.3ms	0.8ms	35.9ms	9.1	6	366	56	11	2,954

Table SM.1 Performance Comparison of Dijkstra, A*, Lazy A*, and Lazy Weighted A* Under Varying λ

An experiment was performed to measure the performance of these different algorithms. Each algorithm was used to plan 5,000 routes between random donor hospitals and transplant centers with departure times drawn between 2015 and 2024. The transport times of the routes computed by the different algorithms were compared, as well as the run times of the algorithms. The results can be found in Table SM.1. For all 5,000 routes, Dijkstra's, A*, and Lazy A* identified the fastest possible route. For Weighted Lazy A*, the percent of routes that were optimal fell as the parameter λ increased.

Dijkstra's algorithm required an average of 174.8 ms to compute a fastest path. A* required 29.4 ms, nearly 6x faster. As expected, this improvement came from a reduction in the number

of nodes that were expanded. The average number of nodes expanded under Dijkstra’s algorithm was 347.5 compared with 37.2 using A*. Lazy A* improved this further with an average run time of 3.7 ms, nearly 8x faster than A*. The two algorithms expanded similar numbers of nodes (37.2 compared to 36.2), however Lazy A* greatly reduced the number of flight database queries. Under A* an average of 8,926 queries were performed versus 268 with Lazy A*. Further reductions in run time were possible with the Lazy Weighted A* variants.

SM.2. Results of Sensitivity and Robustness Analyses

We first provide results of the sensitivity analysis discussed in Section 4.1.3 in Tables SM.2–SM.9.

Table SM.2 Total transportation hours under different assignment constraints. (Historical Estimate: 130,834 hr)

Maximum KDPI Increase	Maximum Additional Wait (weeks)				
	4	6	8	10	12
5	104,176	100,299	97,886	96,251	95,061
10	97,321	94,021	91,973	90,684	89,843
15	93,732	90,950	89,329	88,305	87,625
20	91,774	89,294	87,923	87,031	86,419
25	90,374	88,178	86,948	86,148	85,595

Table SM.3 Shipments greater than 10 hrs under different assignment constraints. (Historical Estimate: 2,066)

Maximum KDPI Increase	Maximum Additional Wait (weeks)				
	4	6	8	10	12
5	929	853	801	762	750
10	808	758	715	689	684
15	742	718	688	665	660
20	735	692	668	626	625
25	709	671	649	615	599

Table SM.4 Shipments greater than 12 hrs under different assignment constraints. (Historical Estimate: 1,308)

Maximum KDPI Increase	Maximum Additional Wait (weeks)				
	4	6	8	10	12
5	521	482	448	420	402
10	463	422	388	376	365
15	423	391	376	366	356
20	405	380	367	351	345
25	401	376	369	341	332

Table SM.5 Shipments less than 1 hr under different assignment constraints. (Historical Estimate: 4,703)

Maximum KDPI Increase	Maximum Additional Wait (weeks)				
	4	6	8	10	12
5	6,934	7,502	7,828	8,112	8,308
10	8,018	8,607	8,968	9,184	9,392
15	8,628	9,191	9,532	9,706	9,857
20	9,040	9,564	9,840	9,982	10,091
25	9,327	9,785	10,014	10,151	10,254

Table SM.6 Air freight shipments under different assignment constraints. (Historical Estimate: 3,325)

Maximum KDPI Increase	Maximum Additional Wait (weeks)				
	4	6	8	10	12
5	2,479	2,348	2,260	2,199	2,176
10	2,233	2,144	2,103	2,066	2,068
15	2,126	2,081	2,031	2,007	1,985
20	2,101	2,038	2,002	1,995	1,960
25	2,054	2,005	1,981	1,958	1,934

Table SM.7 Median transportation distance (nautical miles) with 25th and 75th percentiles under different assignment constraints. (Historical: 129 (41, 220))

Maximum KDPI Increase	Maximum Additional Wait (weeks)				
	4	6	8	10	12
5	81 (14, 174)	72 (12, 164)	67 (11, 158)	63 (10, 154)	60 (9, 149)
10	65 (10, 157)	56 (8, 146)	51 (7, 140)	49 (7, 136)	46 (6, 132)
15	55 (8, 146)	49 (7, 137)	43 (6, 131)	41 (5, 128)	40 (5, 125)
20	51 (7, 139)	43 (6, 131)	40 (5, 127)	39 (4, 122)	37 (4, 120)
25	46 (6, 135)	41 (5, 127)	38 (4, 122)	37 (4, 119)	35 (4, 118)

Table SM.8 Percent of recipients with KDPI increase with median increase of those with an increase under different assignment constraints.

Maximum KDPI Increase	Maximum Additional Wait (weeks)				
	4	6	8	10	12
5	30% (3)	33% (3)	35% (4)	37% (4)	38% (4)
10	36% (6)	39% (6)	41% (6)	42% (7)	43% (7)
15	39% (9)	41% (9)	43% (9)	44% (9)	45% (10)
20	40% (12)	42% (12)	43% (12)	44% (12)	45% (12)
25	40% (14)	42% (14)	43% (14)	44% (14)	45% (15)

Table SM.9 Percent of recipients with waiting time increase with median increase of those with an increase in days under different assignment constraints.

Maximum KDPI Increase	Maximum Additional Wait (weeks)				
	4	6	8	10	12
5	32% (16)	35% (24)	37% (31)	38% (37)	39% (43)
10	38% (16)	41% (24)	42% (31)	43% (37)	43% (43)
15	42% (17)	44% (24)	45% (31)	45% (37)	45% (42)
20	44% (17)	46% (24)	46% (31)	47% (37)	46% (42)
25	46% (17)	47% (25)	47% (31)	47% (38)	47% (42)

Second, we provide the results of the robustness analysis discussed in Section 4.1.4 in Figure SM.1. Solid horizontal lines are estimates of historical values, while those with points are results from the $k = 1, \dots, 10$ optimization problems. For the bottom right plot, the 25th percentile is in blue, the median is in yellow, and the 75th percentile is in green.

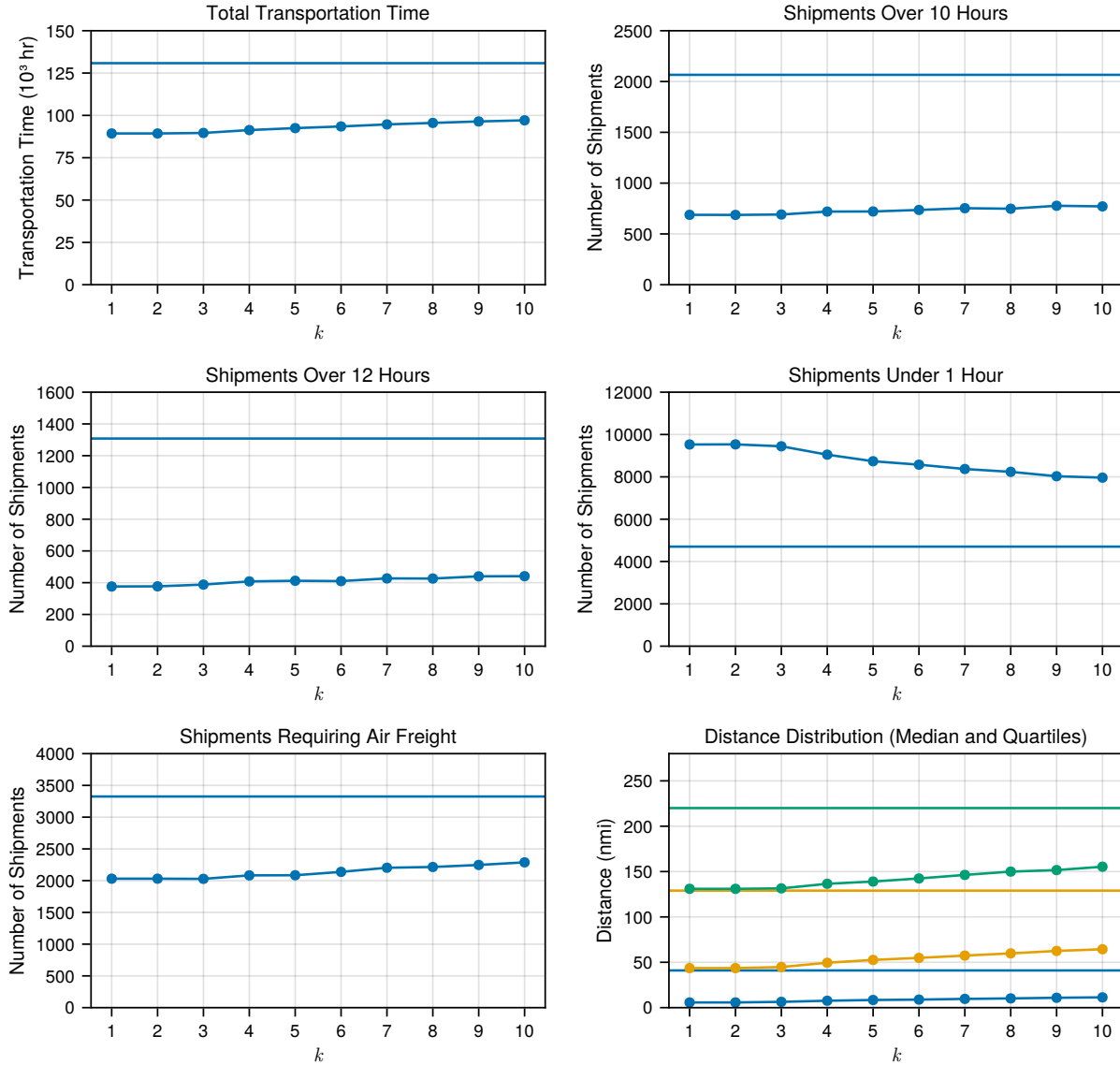


Figure SM.1 (Color online) Results of Robustness Analysis in Section 4.1.4

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